AI-Assisted Single-Cell Diagnostic Prototype

Multimodal Optical Microscopy & High-Precision Electrochemical Analysis

CAUTION — RESEARCH PROTOTYPE: *RESEARCH PROTOTYPE NOTICE: This system is developed strictly for research, educational, and computational evaluation purposes. It is not an FDA-cleared medical device and must not be used as a clinical diagnostic tool or to guide medical treatments.*

Sample ID:	C101P62ThinF_IMG_20150923_170344_cell_61	**Analysis Date:**	2026-10-22 19:01:24
Specimen Type:	Single Erythrocyte (RBC)	**Sensory Setup:**	Optical Brightfield + CV / EIS
Instrumentation:	Potentiostat DAQ (16-bit)	**System Core:**	Multimodal AI Pipeline

Diagnostic Classification Findings

Multimodal Prediction	Confidence	Research Risk Score
HEALTHY	**99.3%**	**0.18 / 1.00**

Multimodal Biomarker Quantitative Extraction

Image Feature	Value
Cell Area	1547.5
Circularity (0-1)	0.3598
Aspect Ratio	1.0806
Texture Variance	1715.54
Mean Intensity	6.99

Electrochemical Feature	Value
Peak Anodic Current (Ipa)	13.05 µA
Charge Transfer Res. (Rct)	1312.49 Ω
Capacitance (Cdl)	1.99 µF
Peak Potential Diff (ΔEp)	70.04 mV
Impedance |Z| at 1 Hz	1353.95 Ω

Analyst Observations

Automated Flask Test